Tutorial - 11

21/4/26

classification

calculating the confusion matrix

Dataset :-

Sample	feature 1 (Age)	feature 2 (Income)	fea 3 (credit score)	Actual class	Predicted Probability
S1	25	40	650	1	0.90
S2	30	60	700	1	0.75
S3	45	35	600	0	0.60
S4	50	30	580	0	0.40
S5	28	55	720	1	0.30

Tasks

1. use threshold 0.5 and compute the confusion matrix
2. calculate the following metrics
   - Accuracy
   - Precision
   - Recall (TPR)
   - f1 - score
3. calculate the false positive Rate (FPR)
4. compute TPR and FPR for thresholds

   0.7, 0.6, 0.5, 0.4, & 0.3 (In exam only one threshold
5. plot the ROC points

class 1 → positive

class 0 → negative

1) Threshold = 0.5

Prob ≥ 0.5 → class 1

prob < 0.5 → class 0

Actual class	Pred Prob	Pred output
1 (P)	0.90	1 (P) (TP)
1	0.75	1 (TP)
0	0.60	1 (FP)
(F) 0	0.40	0 (TN)
1	0.30	0 (F) (FN)

based on threshold we predict our output

TP → True positive

FP → false positive

TN → True negative

FN → false negative

Confusion matrix :

(Y)

	Actual 1	Actual 0
pred 1	TP (2)	FP (1)
pred 0	FN (1)	TN (1)

2×2 (X) (Two classes 1 & 0)

1) Accuracy : $\frac{TP + TN}{\text{Total no.}} = \frac{2+1}{5} = \frac{3}{5} = 0.6$

2) precision : how precise your data is

$$= \frac{TP}{TP+FP} = \frac{2}{2+1} = \frac{2}{3} = 0.67$$

3) Recall : (True positive rate)

$$= \frac{TP}{TP+FN} = \frac{2}{2+1} = 0.67$$

4) f1 Score :

$$= \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} = \frac{2 \times 0.67 \times 0.67}{0.67 + 0.67}$$

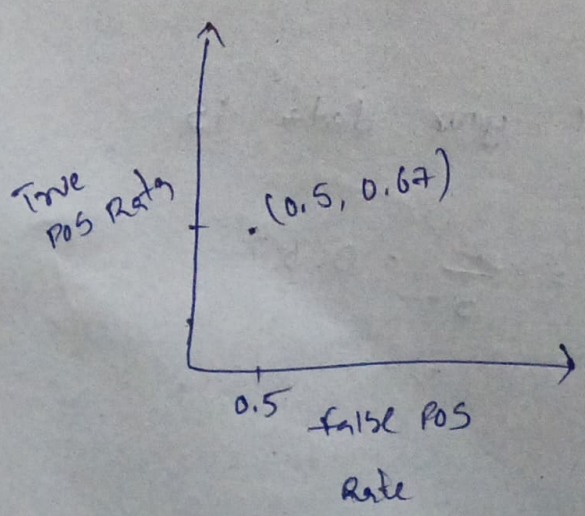
$$= 0.67$$

3) false positive rate :-

(f1)

$$\frac{FP}{FP+TN} = \frac{1}{1+1} = 0.5$$

5) ROC curve :



2) Threshold = 0.7

prob ≥ 0.7 → class 1

prob < 0.7 → class 0

Actual class	pred prob	pred output
1	0.90	1 (TP)
1	0.75	1 (TP)
0	0.60	0 (FN)
0	0.40	0 (TN)
1	0.30	0 (FN)

TP (2)	FP (0)
FN (1)	TN (2)

$$TPR = \frac{TP}{TP+FN} = \frac{2}{2+1} = 0.67$$

$$FPR = \frac{FP}{FP+TN} = \frac{0}{0+2} = 0$$

3) Threshold = 0.6

(2) TP	(1) FP
FN (1)	(1) TN

Act	pred
1	1 (TP)
1	1 (TP)
0	1 (FP)
0	0 (TN)
1	0 (FN)

$$TPR = \frac{TP}{TP+FN} = \frac{2}{2+1} = 0.67$$

$$FPR = \frac{FP}{FP+TN} = \frac{1}{1+1} = 0.5$$